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# MILITARY SPECIFICATION
# CARTRIDGE 40MM, MULTIPLE PROJECTILE, XM576
# LOADING, ASSEMBLING AND PACKING

## 1. SCOPE

1.1 This specification is the applicable detail specification for Cartridge, 40MM, Multiple Projectile, XM576 Loading, Assembling and Packing.

## 2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposals form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

| | |
|---|---|
| MIL-P-116 | - Preservation, Methods of |
| MIL-A-2550 | - Ammunition and Special Weapons, General Specification for |
| MIL-P-46610 | - Primers, Percussion, Styphnate and Chlorate Types, for Small Arms Ammunition |

STANDARDS

Federal

Federal Test Method Standard No. 151 - Metals; Test Methods

Military

| | |
|---|---|
| MIL-STD-105 | - Sampling Procedures and Tables for Inspection by Attributes |
| MIL-STD-109 | - Quality Assurance Terms and Definitions |
| MIL-STD-129 | - Marking for Shipment and Storage |
| MIL-STD-331 | - Fuze and Fuze Components, Environmental and Performance Tests for |
| MIL-STD-639 | - Surface Roughness Values Preferred for Army Ordnance Materiel |
| MIL-STD-1167 | - Ammunition Data Card |
| MIL-STD-1235 | - Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes |

FSC 1310

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2.1 (Cont'd)

DRAWINGS

Munitions Command

| | |
|---|---|
| C10542344 | - Cartridge, 40MM, Multiple Projectile XM576 |
| C8835104 | - Wooden Packing Box |
| C8835105 | - Carton |
| IEL-10542344 | - Index of Inspection Equipment Lists for Cartridge 40MM Multiple Projectile, XM576 Loading, Assembling and Packing |

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General.- Materials, parts and assemblies shall comply with the requirements specified on the applicable drawings, referenced specifications, and the following:

3.2 Plastic defects.- All components and assemblies shall be free from blisters, bubbles, cracks, crazing, mottle, porosity or any other defects.

3.3 Metal defects.- All components and assemblies shall be free from cracks, splits, cold shuts, inclusions, porosity or any other defects.

3.4 Loading.

3.4.1 Propellant.- Propellant shall comply with the requirements of the applicable drawing.

3.4.2 Primer.- Primer shall comply with the requirements of the applicable specification.

3.4.3 Prior to loading, the cartridge case, base plug and propellant cup shall be free of dirt, contamination, oil, water or other defects detrimental in ballistics.

3.4.4 After loading, the cartridge case and base plug shall be free of propellant grains on threads or seating surfaces.



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3.5 Maximum profile.- The maximum profile of the cartridge shall comply with the applicable drawing.

3.6 Functional.

3.6.1 Temperature condition.- Cartridges shall be temperature conditioned at -65, +70 and +155 degrees F. for a minimum of four hours and tested for the following phases:

3.6.2 Recoil momentum.- The cartridges shall function satisfactorily and each of the above temperature conditioned samples shall produce a mean launcher recoil momentum of 2.95 ± 0.15 pound-seconds and a standard deviation ($\sigma$) not exceeding 0.15 pound-seconds.

3.6.3 Target pattern.- The cartridge when fired, shall not fail to release pellets prior to target impact and shall produce a target pattern at 40 meters containing no less than 13 pellets within a 5 foot diameter.

3.6.4 Extraction.- After firing, cartridge cases shall be capable of being extracted from gun chamber manually.

3.6.5 Waterproof test.- The cartridges shall not fail to fire after being subjected to a pressure differential of $7\frac{1}{2}$ pounds per square inch (psi) for 30 seconds.

3.6.6 Velocity.- Velocity shall be recorded for informational purposes only.

3.7 Workmanship.- The requirements for workmanship are as specified on the applicable drawings, referenced specifications and the following:

3.7.1 Burr.- No part shall have a burr which might interfere with the assembly or function of the round, or which might be injurious to personnel handling the item.

3.7.2 Foreign matter.- No part or assembly shall contain dirt, chips, grease, oil, rust, corrosion or other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.



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4.1.1 Quality Assurance terms and definitions.- Reference shall be made to MIL-STD-109 to define Quality Assurance terms used.

4.1.2 Inspection.- Inspection shall be in accordance with MIL-A-2550, except as specified herein.

4.2 First article sample.

4.2.1 Initial production sample.- At the beginning of regular production, a sample shall be submitted in accordance with contract requirements and shall consist of:

115 cartridges, complete

Ninety-five of the above cartridges shall be forwarded to a designated proving ground for ballistic testing. The sample shall be manufactured in the same manner using the same materials, equipment, process and procedures as will be used in regular production. All parts and materials including packaging and packing shall be the same as used for regular production and shall be obtained from the same source of supply.

4.2.1.1 Examination and test.- After inspection and provisional acceptance at source, the sample shall be inspected for all the requirements of the drawings and specifications at a Government laboratory or such other facility specified in the contract.

4.2.1.2 Initial production sample failure.- Failure of the sample to comply with the requirements of the drawings and specifications shall result in sample disapproval.

4.3 Inspection provisions.

4.3.1 Lot.

4.3.1.1 Lot formation.- A lot shall consist of loaded cartridges produced by one manufacturer in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision. Each lot shall contain:

a. Cartridge cases from one interfix lot number from one manufacturer.

b. Projectiles from one interfix lot number from one manufacturer.

c. Primers from one lot number from one manufacturer.

d. Propellant from one lot number from one manufacturer.



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4.3.1.2 Submission of product.- The product shall be submitted in accordance with MIL-STD-105 or MIL-STD-1235, as applicable.

4.3.2 Examination.- One hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed on a defect basis in accordance with the classification of defects contained herein, using the sampling plans contained in MIL-STD-105 or MIL-STD-1235, as applicable. An AQL of 0.25 percent for each major defect and an AQL of 0.40 percent for each minor defect shall be used. All non-conforming material shall be rejected. The Ordnance Standard Finish Blocks prescribed by MIL-STD-639 shall be used as a basis of comparison for surface roughness determination.

4.3.2.1 Cartridge 40MM, XM576 Drawing C10542344.

| | Defect | Method of Inspection |
|---|---|---|
| CRITICAL | | |
| 1. | Primer above flush | Gage |
| 2. | Primer below flush, max. | Gage |
| MAJOR | | |
| 101. | Torque | Torque wrench |
| 102. | Diameter of bourrelet, max. | Gage |
| 103. | Improper assembly | Visual |
| 104. | Adhesive sealant missing or inadequate | Visual |
| 105. | Evidence of primer deterioration | Visual |
| 106. | Material defective | Visual |
| MINOR | | |
| 201. | Length, total, max. | Gage |
| 202. | Base plug not flush | Gage |
| 203. | Crimp for full 360 degrees of case mouth improper | Visual |
| 204. | Marking missing, incorrect or illegible | Visual |
| 205. | Burr | Visual |
| 206. | Foreign matter | Visual |

4.3.2.2 Bandoleer, Drawing F8835107 (before packing in carton)

CRITICAL

None defined

MAJOR

None defined



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4.3.2.2 (Cont'd)

| Defect | Method of Inspection |
|---|---|
| MINOR | |
| 201. Improper number of cartridges | Visual and Manual |
| 202. Plastic support missing | Visual and Manual |
| 203. Bandoleer snaps not closed | Visual |

4.3.2.3 Carton, Drawing C8835105 (before sealing)

CRITICAL

None defined

MAJOR

None defined

| | |
|---|---|
| MINOR | |
| 201. Improper number of bandoleers in carton | Visual |
| 202. Separator missing | Visual |

4.3.2.4 Carton, Drawing C8835105 (prior to sealing bag)

CRITICAL

None defined

MAJOR

None defined

| | |
|---|---|
| MINOR | |
| 201. Container damaged | Visual |
| 202. Corners not blunted | Visual |
| 203. Contents loose | Visual |
| 204. Marking missing, incorrect or illegible | Visual |

4.3.2.5 Wooden packing box, Drawing C8835104 (before sealing)

CRITICAL

None defined

MAJOR

None defined



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4.3.2.5 (Cont'd)

| Defect | Method of Inspection |
|---|---|
| MINOR | |
| 201. Carton facing wrong way | Visual |
| 202. Marking of bag missing, incorrect or illegible | Visual |
| 203. Bag torn or perforated | Visual |
| 204. Bag improperly sealed | Visual |

4.3.2.6 Wooden packing box, Drawing C8835104 (after sealing)

CRITICAL

None defined

MAJOR

None defined

MINOR

| | |
|---|---|
| 201. Box damaged | Visual |
| 202. Contents loose or broken | Visual |
| 203. Car seal missing or improperly positioned | Visual |
| 204. Marking missing, incorrect or illegible | Visual |

4.3.3 Maximum profile gage.- Each cartridge shall be gaged in a Government certified maximum profile gage. Failure of the assembly to comply with 3.6 shall be cause for rejection.

4.3.4 Inspection testing.- The following tests shall be performed in accordance with 4.4.

4.3.4.1 Propellant weight.- Each propellant charge shall be weighed to determine compliance with the requirements of the applicable drawing. From the propellant charges that comply, one out of every ten shall be check weighed. If any charge, being check weighed fails to comply with the requirements of the applicable drawing, each charge shall be 100 percent check weighed until 190 consecutive charges have complied. When 190 consecutive check weighed charges have complied, the sampling plan previously used may be implemented. Propellant charges that do not comply shall be rejected.



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4.3.4.2 Security of projectile crimp.- Twenty cartridges from each eight (8) hours production or fraction thereof will be subjected to this test. Failure of any cartridge to comply with the requirements of the applicable drawing shall be cause for rejection of the lot.

4.3.4.3 Heat seal test of packing seals.- The sample size and acceptance criteria shall be as specified in the applicable specification.

4.3.5 Functional.- The following proving ground tests will be performed by the Government.

4.3.5.1 First article sample.

4.3.5.1.1 Initial production sample.- The quantities listed in Table I will be ballistically tested at temperatures indicated and the results of the test will be evaluated in accordance with the criteria set forth in Table III, except no retest will be permitted.

TABLE I

| Temperature | Quantity |
|---|---|
| -65 degrees F. ± 5 degrees F. | 30 |
| +70 degrees F. ± 5 degrees F. | 30 |
| +155 degrees F. ± 5 degrees F. | 30 |

4.3.5.2 Production lot sample.- The quantities listed in Table II will be ballistically tested at temperatures indicated and the results of the test will be evaluated in accordance with the criteria set forth in Table III.

TABLE II

| Temperature | Quantity |
|---|---|
| -65 degrees F. ± 5 degrees F. | 20 |
| +70 degrees F. ± 5 degrees F. | 20 |
| +155 degrees F. ± 5 degrees F. | 20 |

TABLE III

| Defective Characteristic | Accept | Reject |
|---|---|---|
| Primer remains in pocket but is physically loose | 10% or less | Over 10% |
| Failure to extract by spring operation of extractor | 10% or less | Over 10% |
| Failure to extract by hand operation of extractor | 0 | 1 |
| Primer falls out of pocket | 0 | 1 |
| Blown primer | 0 | 1 |



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4.3.5.2 (Cont'd)

TABLE III (Cont'd)

| Defective Characteristic | Accept | Reject |
|---|---|---|
| Misfire | 0 | 1 |
| Target pattern | 2 | 3 |
| Recoil momentum 1/ | | |
| Velocity (information only) | | |

1/ The lot shall be rejected if the mean recoil momentum or standard deviation fails to comply with 3.6.2.

4.3.5.2.1 Suitability assurance test.- Beginning with the first lot produced by each producer and continuing until that producer has submitted three consecutive lots that are accepted in accordance with Table III, an additional sample of ninety cartridges shall be selected from each lot and ballistically tested at temperatures indicated in Table I and the results of the test will be evaluated in accordance with criteria set forth in Table III.

4.3.5.3 Waterproof test.- Five cartridges from the initial production lot and five cartridges from each production lot shall be ballistically tested to determine compliance with 3.6.5. Failure of any cartridge to comply shall be cause for rejection of the lot represented by the sample.

4.3.6 Inspection equipment.- Index of Inspection Equipment List No. IEL-10542344 identifies the inspection equipment required to perform the examination and tests prescribed herein. Equipment design shall be in accordance with code designation on the applicable IEL. (Reference is made to Inspection Equipment List for definition of coding format.)

4.4 Test methods and procedures.

4.4.1 Propellant weight.- Check weighing shall be accomplished independently of the original weighing, using a difference balance from that used to make the weighing. The balance shall be calibrated at the beginning of each working day.

4.4.2 Security of projectile crimp.- The samples supplied in accordance with 4.3.4.2 shall be mounted in an approved pull test device in such a manner that the cartridge case will be held stationary. The force will be applied to the sabot until it is separated from the case. Force required to achieve separation must be within specified limits. This is not a destructive test. Parts may be returned to the lot after re-assembly, re-crimping of case mouth and re-application of adhesive sealant.



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4.4.3 Torque.

4.4.3.1 Base plug to cartridge case.- The cartridge case assemblies shall be mounted in a suitable holding device in such a manner that the cartridge case is prevented from turning. The applicable torque shall be applied to the base plug in the direction of assembly. Testing shall be conducted while thread sealant is wet.

4.4.4 Heat seal test of packing seals.- The procedure for this test shall be in accordance with MIL-P-116. Seals so tested shall not be returned to the lot.

4.4.5 Functional.

4.4.5.1 Ballistic test.- The cartridges supplied in accordance with 4.3.5.1 or 4.3.5.2 will be fired from an M-79 Launcher mounted on a recoil measuring pendulum. The gun is fired by means of a spring trip. Firing the gun causes the pendulum to recoil and by means of a stylus attached to the pendulum, the length of swing is marked on a curved surface beneath it. This recoil distance is converted to momentum (lb-sec) using total weight of weapon and pendulum and the period of the pendulum (time required for pendulum to swing through one complete cycle). The mean recoil momentum of the lot being tested shall be corrected by the correction factor determined for the particular launcher being used on the day of the test firing. When recoil values are suspected to be outliers, they shall be tested using the statistical procedure approved by the Government proving ground and the mean recoil momentum and standard deviation adjusted accordingly. Velocity shall be measured at 7.5 feet from the muzzle of the launcher using break screens placed 5 feet and 10 feet from the muzzle. A target measuring 8 feet by 8 feet shall be placed 40 meters from the muzzle of the launcher. Observations shall be made for the defective characteristics listed in Table III and recorded for each round ballistically tested.

4.4.5.1.1 Suitability assurance test.- From the additional samples selected in accordance with 4.3.5.2.1, the cartridges shall be packaged in their regular shipping containers and subjected to transportation vibration environment in accordance with MIL-STD-331, test number 104. After transportation vibration, the cartridges shall be unpacked (and de-linked if necessary) and subjected to the JAN temperature and humidity cycle (14 days) specified in MIL-STD-331, test number 105. Following the temperature and humidity cycle, the cartridges shall be temperature conditioned in accordance with Table I. The sample of ninety cartridges will be fired as specified in 4.4.5.1.



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4.4.5.2 Waterproof test.- The samples provided in accordance with 4.3.5.3 shall be temperature conditioned at +70 degrees F. and placed in a container under a 3 inch head of water. The container shall be evacuated to a pressure of 7½ pounds per square inch (15 inches of mercury) below atmospheric pressure and held at that pressure for 30 seconds under a 3 inch head of water. The container shall then be returned to atmospheric pressure and allowed to stand for 30 seconds after which the rounds will be dried and fired within four hours.

## 5. PREPARATION FOR DELIVERY

5.1 Packing, Level A (worldwide shipment).- The cartridges shall be packed in accordance with Drawings C8835104 and C8835105.

5.2 Marking and labeling.- Packing boxes shall be marked and labeled in accordance with the applicable drawings cited in 5.1.

5.3 Data card.- Data card information shall be as specified in MIL-STD-1167.

## 6. NOTES

6.1 Visual examination qualification.- When compliance with the applicable requirements is in doubt as a result of visual examination, this characteristic may be measured or gaged to determine acceptability.

6.2 Ordering data.- Procurement documents should specify the title, number, and date of this specification.

Custodian:

Army - MU

Preparing activity:

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